Sahaj Bhadja

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Summary

Forward Deployed Software Engineer with 2+ years translating ambiguous requirements into **production AI**, data, and application solutions using **Python**, TypeScript, cloud platforms, **REST APIs**, automated testing, observability, and CI/CD.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Software Engineering, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Data Retrieval, Operating Systems, Software Engineering

Experience

USC Dornsife **September 2025 – Present**
Forward Deployed Software Engineer *Los Angeles, California*

- Architected a multi-tenant platform using Azure SQL and Blob Storage with row-level security and normalized schemas for 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds.
- Built FastAPI integrations across PostgreSQL, SQL Server, and cloud storage, processing 5K+ monthly jobs with validation, reconciliation, error handling, and automated CI/CD releases sustaining 99%+ production reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, California*

- Designed concurrent processing and secure multi-format ingestion through REST APIs and AWS RDS, executing 1K+ files per batch with fault-tolerant recovery, usage tracking, and stable performance under load.
- Built Python services and responsive React interfaces for uploads, history, configuration, and live feedback, reducing user turnaround 60% while supporting reliable production workflows used by 150+ instructors.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer, SecureSync *India*

- Designed object-oriented API integrations and AWS Glue transformations for JSON, CSV, and Parquet data, improving response times 30% and pipeline efficiency 68% while reducing technical debt 20%.
- Automated nightly builds, installations, unit tests, integration tests, and reports through Jenkins, validating reliable cross-platform synchronization while collaborating directly with Microsoft and GitHub stakeholders throughout Agile delivery.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern, VibrantChem *India*

- Implemented eight MERN microservices for authentication, permissions, billing, resources, customers, leads, and notifications, reliably synchronizing 2M+ records through REST APIs and event-driven WebSocket updates across production workflows.
- Built responsive React workflows and automated Docker deployments through Jenkins CI/CD, achieving 95%+ build validation while integrating operational monitoring and analytics across controlled staging and production environments.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built cross-platform mobile and web booking workflows with Flask APIs, Firebase synchronization, payments, and webhooks, processing hourly Databricks micro-batches across 50K+ records with consistently reliable sub-200-millisecond responses.
- Designed end-to-end workflows, mapped 12+ integration points, and presented technical solutions to client stakeholders, reducing design errors 25% while supporting effective collaborative cross-functional implementation planning and alignment.

Skills

Engineering & AI: Python, JavaScript, TypeScript, Java, PyTorch, Transformers, LLM Systems, FAISS, Feature Engineering
Cloud & Data: Azure, AWS, GCP, PostgreSQL, SQL Server, MongoDB, ETL Pipelines, Data Modeling, Data Validation
Applications: React, Node.js, FastAPI, Flask, REST APIs, Microservices, WebSockets, Authentication
Delivery: Docker, Kubernetes, Jenkins, CI/CD, Git, GitHub, Unit Testing, Integration Testing, Observability, Debugging, Security, Documentation, Requirements Discovery, Stakeholder Communication

Projects

Big-Brain Live | *LLaMA-2, FAISS, PostgreSQL, GCP, REST APIs*

- Engineered a reliable cloud-based data ingestion and indexing pipeline processing 50K+ records, implementing incremental updates, metadata persistence, and performance tuning to reduce retrieval latency by 40% overall.

HMIS - Hospital Management Information System Live | *TypeScript, MERN, REST APIs, Twilio, Sentry*

- Implemented consent-based access, responsive operational workflows, REST integrations, automated alerts, and Sentry monitoring through scalable TypeScript services supporting 100+ daily operations while reducing administrative workload by 30%.

FinNexus Live | *TypeScript, Plaid, MongoDB, WebSockets, Kubernetes*

- Built a fintech platform supporting 50K+ users and 1M+ transactions with Plaid integration, secure APIs, real-time synchronization, advanced filtering, and reliable Kubernetes-based deployment workflows at production scale.